

Mathematical Abstract of a Two-Channel Watermarking System

(DCT–QIM + Fourier–Mellin) with RS Coding and Soft Beam Decoding

Implementation: `trajectory_watermark_2ch.cpp` (v2.0 – `--strong` flag, geometric realignment, temporal aggregation, multi-key management; images and video)

Abstract

This note gives a concise mathematical description of a *two-channel* watermarking pipeline. **Channel A** embeds a short readable payload (up to 13 characters) into the mid-frequency DCT coefficients of 8×8 image blocks using Quantization Index Modulation (QIM), protects the payload with Reed–Solomon (RS) channel coding over $\mathbb{F}_{2^8}$, and recovers candidates via soft symbol scoring and a byte-level beam search. **Channel B** embeds a 16-bit *fingerprint* (the first 16 bits of SHA-256(message)) into the magnitude of the discrete Fourier transform on concentric rings (Fourier–Mellin domain), together with a keyed synchronization template; rotation, scaling and translation of the host image become an angular shift, a radial shift and no effect on the magnitude, respectively, so the fingerprint survives geometric attacks that destroy Channel A. Channel B is optional and controlled by the `--strong[=X]` flag (strength X , default 1.5): when the flag is absent only Channel A is embedded, leaving the image essentially undistorted. A case-insensitive normalized Levenshtein similarity reports a match percentage between a user-supplied ideal string and the recovered candidate.

1 Signal model

Let $I : \Omega \subset \mathbb{Z}^2 \rightarrow [0, 255]$ denote the luminance component Y of an image (or of a single video frame). Partition I into non-overlapping 8×8 blocks $B_{m,n}$. For each block compute the 2-D Discrete Cosine Transform (DCT):

$$C_{u,v}^{(m,n)} = \frac{\alpha(u)\alpha(v)}{4} \sum_{x=0}^7 \sum_{y=0}^7 I_{m,n}(x,y) \cos \frac{(2x+1)u\pi}{16} \cos \frac{(2y+1)v\pi}{16},$$

where

$$\alpha(k) = \begin{cases} 1/\sqrt{2}, & k = 0, \\ 1, & k \geq 1. \end{cases}$$

Each block carries *two* mid-frequency coefficients of interest (one symbol embedded redundantly in both):

$$c_1 = C_{u_1^*, v_1^*}^{(m,n)}, \quad (u_1^*, v_1^*) = (2, 2) \text{ (zigzag 12)}, \quad c_2 = C_{u_2^*, v_2^*}^{(m,n)}, \quad (u_2^*, v_2^*) = (3, 3) \text{ (zigzag 24)}.$$

2 Payload coding and symbol mapping

Let the user payload be a short ASCII string M with $L = |M| \leq 13$ characters. A 16-byte descriptor is formed as

$$P_0 = R, \quad P_1 = L, \quad P_{2+i} = M_i \ (i = 0, \dots, L-1), \quad P_{15} = \text{CRC8}(P_2, \dots, P_{L+1}),$$

where R is the replication factor (self-describing: the decoder reads it back from the watermark). Reed–Solomon encoding over $\mathbb{F}_{2^8}$ with $n_{\text{sym}} = 16$ parity symbols expands P into a codeword

$$E = (e_0, e_1, \dots, e_{N-1}) \in \{0, \dots, 255\}^N, \quad N = 32 \text{ (16 data bytes + 16 parity bytes)}.$$

Each byte e_i is expanded into four base-4 symbols (MSB-first):

$$e_i \mapsto (s_{4i}, s_{4i+1}, s_{4i+2}, s_{4i+3}), \quad s_j \in \{0, 1, 2, 3\},$$

yielding a symbol sequence $\{s_k\}_{k=0}^{127}$ of length $K = 128$.

Key-derived slot allocation. A master key κ seeds a deterministic Fisher–Yates shuffle of all 8×8 blocks using a 32-bit linear congruential generator whose seed is the first 32 bits of $\text{SHA-256}(\kappa)$. The first $4 + 128R$ blocks of the shuffled list are the watermark *slots*: 4 pilot slots followed by $128R$ payload slots. The key is read from `masterkey.txt` or the environment variables `TW_MASTER_KEY` / `TW_PREV_KEYS`.

3 QIM embedding (Channel A)

Quantization Index Modulation embeds a symbol $s \in \{0, \dots, V-1\}$ into a coefficient c by shifting c to the nearest centre of a uniform partition with step Δ and V sublevels:

$$\text{embed}(c; s) = \left\lfloor \frac{c}{\Delta} \right\rfloor \Delta + \left(s + \frac{1}{2}\right) \frac{\Delta}{V}, \quad \Delta = 100, V = 4.$$

In the implementation each symbol s_k is embedded into *both* coefficients c_1, c_2 of R spatially distributed blocks ($R = 18$ by default, configurable with `--r_rep`); each block therefore contributes two votes per slot. Additionally, 4 pilot slots receive strong low-frequency energy (DC scaled by 4, coefficients $(0, 1), (1, 0)$ by 2, $(1, 1)$ by 1, all scaled by pilot strength 80) for synchronization and coarse detection. After modifying the coefficients, the inverse DCT reconstructs the spatial samples, clamped to $[0, 255]$.

Saturated blocks (known limitation). The QIM pattern is embedded in the luminance and its spatial swing (at most ± 19 grey levels for the strongest symbol) is applied through the YCbCr→RGB round-trip with clamping to $[0, 255]$. Blocks whose pixels are pure black, pure white or fully colour-saturated (any of R, G, B at 0 or 255) therefore cannot carry the pattern: the clamp destroys the QIM constellation and those blocks decode to garbage. Natural photographs rarely contain such blocks (the scheme recovers the payload at 100% on them), whereas synthetic saturated patterns (e.g. the `testsrc/smpTEbars` ffmpeg sources, solid black or white images) fail by design. For functional testing a real photo or a grey-plus-noise host should be used.

4 Extraction of Channel A: soft scoring and beam search

For each slot the algorithm inspects its R blocks and, for each of the two coefficients, estimates the symbol whose QIM centre is closest to the observed value. Aggregating the $2R$ estimates yields soft frequencies, normalized to soft probabilities

$$p_k(s) = \Pr(\text{symbol} = s \mid \text{observations for slot } k), \quad s \in \{0, 1, 2, 3\}, \quad \sum_s p_k(s) = 1.$$

Symbols are grouped in quadruples (MSB-first) to form candidate bytes, scored multiplicatively

$$\text{score}(\text{byte}) \propto \prod_{j=0}^3 p_{k+j}(s_{k+j}).$$

A beam search of width $W = 64$ keeps, at each of the $N = 32$ byte positions, the top W partial sequences obtained from the top-3 symbol candidates per slot. Final candidate byte sequences

of length N are submitted to Reed–Solomon decoding (Berlekamp–Massey error locator, Chien search for roots, Forney formula for magnitudes). If RS decoding succeeds, a CRC8 check on the payload validates integrity. If no beam candidate decodes, a fallback uses majority voting per slot followed by MSB-first unpacking to produce a best-effort candidate.

5 Channel B: Fourier–Mellin fingerprint (robustness to geometric attacks)

Channel A is block-synchronous and therefore fragile to rotation, scaling and cropping. Channel B adds a low-capacity, RST-invariant payload. Let $\mathcal{F}$ be the DFT of the (zero-padded, power-of-two $N \times N$) luminance and define the log-magnitude

$$\mathcal{M}(u, v) = \log(1 + |\mathcal{F}(u, v)|).$$

Rotation by θ rotates $\mathcal{M}$ (an angular shift $\varphi = \theta$); scaling by σ scales radii ($\rho \rightarrow \sigma\rho$); translation affects only the phase and leaves $\mathcal{M}$ unchanged.

5.1 Synchronization template

A keyed pseudo-random template $T = (T_0, \dots, T_{M-1}) \in \{\pm 1\}^M$ ($M = 256$, LCG seeded with the first 32 bits of SHA-256(κ), with $T_{k+M/2} = T_k$) is embedded *multiplicatively* on $R_s = 8$ low-frequency rings of radii $\rho_j = N(r_0 + (r_1 - r_0) \frac{j}{R_s - 1})$, $r_0 = 0.04$, $r_1 = 0.14$:

$$\tilde{\mathcal{F}}(u, v) = \mathcal{F}(u, v) \cdot e^{GT_k}, \quad (u, v) \text{ on ring } j \text{ at angle } \theta_k = \frac{2\pi k}{M},$$

applied to $k = 0, \dots, M/2 - 1$ and mirrored at the Hermitian-conjugate points $(N-u, N-v)$. Here G is the strength parameter ($G = 1.5$ by default).

5.2 Fingerprint embedding

The fingerprint is the 16-bit integer

$$\text{fp} = \sum_{i=0}^{15} b_i 2^{15-i}, \quad (b_0, \dots, b_{15}) = \text{first 16 bits of SHA-256}(M).$$

It is embedded multiplicatively on 32 higher-frequency rings of radii $\rho'_j = N(0.16 + 0.24 \frac{j}{31})$, $j = 0, \dots, 31$. Each bit b_i is spread over $K = 8$ consecutive angular samples $\theta = \frac{2\pi(iK+k)}{M}$, $k = 0, \dots, K-1$, with factor

$$e^{Gs_i}, \quad s_i = +1 \text{ if } b_i = 1, \quad s_i = -1 \text{ if } b_i = 0,$$

again mirrored on the Hermitian-conjugate points.

5.3 Geometric recovery

Given a possibly attacked image, the detector searches for the rotation and scale that realign the sync template. For a candidate (φ, σ) it samples ring j at radius $\rho_j\sigma$ and angular offset φ (bilinear interpolation, M samples), applies an angular high-pass (subtraction of a circular moving average, window 16), and computes the normalized correlation with T , averaged over the R_s rings:

$$s(\varphi, \sigma) = \frac{1}{R_s} \sum_j \frac{\langle r_j(\varphi, \sigma), T \rangle}{\|r_j\| \|T\|}.$$

A coarse grid scan (85 scale factors $\sigma \in [0.4, 2.5] \times 128$ angular offsets $\varphi \in \{2\pi k/M\}_{k=0}^{M/2-1}$) is followed by a fine refinement ($\pm 3^\circ$ at 0.1° steps, scale $\pm 3\%$). Geometry is considered recovered when the best score exceeds 2.0; the estimate $(\hat{\theta}, \hat{\sigma})$ is returned.

5.4 Fingerprint decoding

At the recovered geometry, the detector re-samples the fingerprint rings, applies an angular high-pass (window 64), and accumulates per-bit energy

$$\text{acc}_i = \sum_{j=0}^{31} \sum_{k=0}^{K-1} r_j[iK + k], \quad \hat{b}_i = [\text{acc}_i > 0].$$

The recovered fingerprint is compared bitwise with the ideal fingerprint (first 16 bits of SHA-256 of the user-supplied ideal string). With $m = \#\{i : \hat{b}_i = b_i\}$ the decision rule is

$$m \geq 14 \Rightarrow \text{PRESENT}, \quad 12 \leq m < 14 \Rightarrow \text{LIKELY}, \quad m < 12 \Rightarrow \text{absent}.$$

6 Geometric realignment of Channel A (v2.0)

When `--strong` is active and the geometric recovery of Section 5.3 returns an estimate $(\hat{\theta}, \hat{\sigma})$ with sufficient confidence, the detector attempts to undo the attack on the *spatial* domain so that the block-synchronous Channel A can be decoded again. Because the synchronization template is π -periodic the rotation has a 180° ambiguity, and the recovered phase does not fix the direction of the rotation, so the detector builds the four rotation candidates

$$\hat{\theta}_c \in \{ \hat{\theta}, \hat{\theta} + 180^\circ, -\hat{\theta}, -\hat{\theta} + 180^\circ \} \pmod{360^\circ}$$

and, for each candidate, re-samples the source image at

$$p = C + \frac{1}{\hat{\sigma}} R_{\hat{\theta}_c}(q - C), \quad C = \text{image centre}, \quad q = \text{destination pixel},$$

which reduces to a pure rotation $p = C + R_{\hat{\theta}_c}(q - C)$ when $\hat{\sigma} \approx 1$. The source image is re-sampled with *nearest-neighbour* interpolation: bilinear filtering would blur the mid-frequency DCT coefficients and destroy the QIM signal, whereas nearest-neighbour sampling preserves them (pixels map back to original samples for rotations that preserve the grid, e.g. 180°). Each candidate is decoded with Channel A; the first attempt that passes the CRC check reports the readable message, otherwise the fingerprint fallback of Channel B applies. In practice the realigned decode succeeds for attacks that preserve the pixel values (e.g. 180° , and $90^\circ/270^\circ$ on near-square images), while attacks that re-sampled with an interpolating filter (e.g. arbitrary-angle rotations that blur the host) reduce to the fingerprint-only fallback.

7 Temporal aggregation for video (v2.0)

Video watermarking marks one frame every $N+1$ (the `--frame` option), so a long extraction pass may contain mostly unmarked frames. Instead of decoding every frame independently, the extractor first computes, for each frame, a *pilot score* equal to the energy of the 4 pilot slots' boosted DCT coefficients $(0, 1), (1, 0), (1, 1)$:

$$\text{pilot}(F) = \sum_{i=1}^4 \left(|C_{0,1}^{(i)}| + |C_{1,0}^{(i)}| + |C_{1,1}^{(i)}| \right).$$

Watermarked frames score far above unmarked ones. Frames whose score exceeds the midpoint $\frac{1}{2}(\text{mean} + \text{max})$ are selected, their luminance planes are averaged, and a *single* Channel A decode is performed on the average:

$$\bar{Y} = \frac{1}{|\mathcal{S}|} \sum_{F \in \mathcal{S}} Y_F, \quad \mathcal{S} = \{\text{selected frames}\}.$$

Because the same key places the same symbols in the same slots on every frame, the watermark adds coherently while content differences and compression noise average out. If the aggregated decode fails (e.g. no pilot detected), the extractor falls back to the per-frame best-of- N scan.

8 Key management (v2.0)

The master key κ remains the verification secret. v2.0 adds a numbered key list stored in `masterkeys.txt`; each entry shows its index, the first 8 hex digits of $\text{SHA-256}(\kappa)$ (for recognition without revealing the key) and an (`in use`) marker for the active key. The commands `--mkwrite`, `--mkread`, `--mkshow` and `--mkset` add, list, show and switch keys, and may be combined with `embed/extract` in a single invocation (executed in command order). Duplicate keys are not added (the existing index is reused). Extraction tries only the active key and `TW_PREV_KEYS`; it never scans the whole list automatically.

Security model. The slot allocation is fully determined by κ (Section 3), so without the key the watermark can be neither located, nor read, nor removed surgically. A second watermark embedded with a *different* key lands on a different, mostly disjoint set of blocks and therefore does *not* erase the original pattern: in experiments the original message remained 100% readable after re-watermarking at low redundancy, while a second embed at the same (or higher) `--r_rep` corrupts $\approx 15\%$ of the original's blocks through block collisions — enough to make the original message unreadable. That is interference (jamming) rather than surgical removal. Knowing the key conversely allows both reading and clean removal, so the key remains the only element protecting the mark.

9 Levenshtein similarity and match percentage

Let s_{ideal} be the user-provided ideal string and s_{rec} the recovered candidate. Compute the Levenshtein distance $d_L(s_{\text{ideal}}, s_{\text{rec}})$ and normalize it to a similarity score

$$\text{sim}(s_{\text{ideal}}, s_{\text{rec}}) = 1 - \frac{d_L(s_{\text{ideal}}, s_{\text{rec}})}{\max\{|s_{\text{ideal}}|, |s_{\text{rec}}|\}},$$

with $\text{sim} = 1$ when both strings are empty. The reported match percentage is

$$\text{match\%} = 100 \cdot \text{sim}(s_{\text{ideal}}, s_{\text{rec}}),$$

formatted with one decimal digit. The comparison is case-insensitive.

10 Two-channel interpretation and the `--strong` flag

- **Channel A** (always active) provides high capacity (a readable message) and strong robustness to lossy compression and noise, but is destroyed by rotation, scaling or cropping, which break the block grid.
- **Channel B** (`--strong[=X]`) provides low capacity (a 16-bit fingerprint) but is RST-invariant: rotation becomes an angular shift, scaling a radial shift, translation has no effect. It enables presence verification after geometric attacks, even when the readable message can no longer be recovered.
- **Default behaviour:** without `--strong` only Channel A is embedded, so the host image is not degraded by the Fourier–Mellin perturbation. With `--strong` the strength X (default 1.5) scales both the sync template and the fingerprint. Larger X increases robustness against geometric attacks at the cost of visible distortion; smaller X keeps the image clean but reduces robustness (typical range 1.0–2.5).
- **Extraction order (v2.0):** the decoder first attempts Channel A; on CRC success it prints the readable message and the match percentage. Otherwise, if (and only if)

`--strong` was given, it recovers the geometry $(\hat{\theta}, \hat{\sigma})$ and, before falling back to the fingerprint, *realigns the image and retries Channel A* (see Section 6). Only if the realigned decode fails does it decode the fingerprint and report the recovered 16 bits, the full SHA-256 of the ideal string and the number of matching bits.

Configuration	Channels embedded	Geometric robustness	Image quality
default (no <code>--strong</code>)	A only	none	high (≈ 38 dB PSNR)
<code>--strong=1.5</code>	A + B	rotation/scale/crop	slightly reduced (≈ 32 dB)

11 Command-line interface and platform notes

- **Embed:** `tw embed --in in.png --out out.png --msg "text" [--r_rep N] [--frame N] [--strong[=X]] [--no_sync]`; the same syntax applies to video files (`.mp4/.mov/.avi/.mkv`), which are watermarked frame-by-frame through `ffmpeg` (one frame every $N+1$ is marked by default).
- **Extract:** `tw extract --in file --profile auto|low|high --ideal "expected" [--frames N] [--r_rep N] [--strong[=X]] [--no_sync]`.
- **Replication factor `--r_rep`:** $4 + 128R$ blocks are required. Small images need a small R (e.g. $R = 1$ for a 128×96 host), large images can use a larger R for more redundancy. If the host has fewer than $4 + 128R$ blocks, the same blocks are *reused (overwritten)*: the payload is corrupted and the image is visibly degraded (measured ≈ 17 dB PSNR for a 128×96 host at $R = 18$ versus ≈ 29 dB at $R = 1$); hosts with enough blocks (e.g. 800×600 at $R = 18$, ≈ 38 dB) are essentially unaffected. Minimum square hosts that avoid reuse: 96×96 ($R=1$), 264×264 ($R=8$), 392×392 ($R=18$, default), 520×520 ($R=32$). Extraction reads R back from the watermark and auto-detects it in $R \in [1, 32]$; for $R > 32$ the same `--r_rep` value should be passed to `extract`.
- **Key management:** a master key is requested at first run and stored in `masterkey.txt`; previous keys can be supplied through `TW_PREV_KEYS` (semicolon-separated) so that old files remain readable. v2.0 adds the numbered `masterkeys.txt` list with the `--mkwrite/--mkread/--mkshow/--mkset` commands (Section 8).
- **Version and help:** `--version` (or `-v`) prints `trajectory_watermark v2.0`; `--help` (or `-h`) prints the English parameter help.
- **Temporal aggregation:** for video extraction, `--frames N` now selects the watermarked frames by pilot score, averages them and decodes once (Section 7), with a best-of- N per-frame fallback.
- **Portability:** self-contained SHA-256, CRC8 and Reed–Solomon implementations (no external crypto libraries); images are read/written through the public-domain single-header `stb_image / stb_image_write`. Builds are provided for Linux (`g++ -O2 -std=c++17`) and Windows (MinGW-w64, fully static `.exe`).

Summary

The described pipeline integrates DCT-domain QIM embedding with RS channel coding and soft decoding with byte-level beam search (Channel A), and a keyed Fourier–Mellin fingerprint with a synchronization template (Channel B), to maximize recoverability of short payloads and to provide geometric-attack robustness that a purely block-based scheme cannot offer. The normalized Levenshtein-based match percentage remains the quantitative, case-insensitive

indicator of correspondence between the ideal and recovered strings, while the 16-bit fingerprint match count provides the decision statistic for RST-attacked content. The `--strong` flag lets the user trade image fidelity against geometric robustness, and keeps the host image undistorted by default.